Chapter 92 — Baseline Profiles & R8 Optimization

> **Part X — Performance, Observability & Quality** · Complexity: **Standard** (≥25 interview questions)
>
> **Prerequisites:** Ch 88 (cold start), Ch 90 (jank), Ch 85 (perf monitoring/Macrobenchmark), Ch 45 (startup). Pairs with Ch 89 (memory), Ch 93 (lists), Ch 41 (release).
>
> **Reference surface in StreamCart:** making the app **fast from the first launch** and **smaller** — **Baseline Profiles** (AOT-compile the startup + critical-path code so it isn't interpreted/JIT-cold, improving **cold start** Ch 88 and **first-scroll jank** Ch 90) and **R8** (shrink, optimize, obfuscate — smaller APK, faster load, dead-code removal) — verified with **Macrobenchmark** (`CompilationMode.None` vs `Partial`) and the field (Ch 85).

92.1 Problem Definition and Real-World Context

On a fresh install (or after an update/profile reset), Android runs app code **interpreted**, then **JIT-compiles** hot methods over time — so the **first** launches and **first** scrolls are **slow/janky** until the JIT warms up. **Baseline Profiles** fix this: a profile lists the **startup + critical-path methods/classes**, and ART **AOT-compiles** them **at install** → the app is **fast from the very first launch** (commonly **20–30%** faster cold start, Ch 88, and smoother first-scroll, Ch 90). Separately, **R8** (Android's shrinker/optimizer/obfuscator) **removes dead code, optimizes, and shrinks** the APK/DEX → **smaller download, faster load, less to compile/initialize**. Together, **Baseline Profiles + R8** are among the **highest-impact, lowest-effort** performance wins — and both are **verified with Macrobenchmark** (Ch 85) so you know the gain is real. This chapter is the deep-dive counterpart referenced by Ch 88 (cold start) and Ch 90 (jank).

Concretely, teams hit:

- **JIT-cold first launches (the core problem).** Without a baseline profile, the **first few launches** run **interpreted → JIT**, so they're measurably **slower** than warmed-up runs. Users judge the app on the **first** impression (Ch 88). A **Baseline Profile** AOT-compiles the startup path so it's fast **immediately**.
- **First-scroll jank (Ch 90).** The first scroll through a list/feed runs **JIT-cold** Compose/ scroll code → **janky** until warm. Including the **scroll path** in the baseline profile makes it **smooth from the start** (Ch 90).
- **No profile / wrong profile.** Shipping without a baseline profile leaves performance **on the table**; a profile that **doesn't cover** the real critical journeys (only launch, not the key screens) under-delivers. The profile must cover **launch + the most-used journeys**.
- **Profile not applied / verified.** A baseline profile must be **packaged** correctly (via the **Baseline Profile Gradle plugin** / `ProfileInstaller`) and **verified** to actually apply — otherwise you think you have it but don't. **Macrobenchmark** (`CompilationMode.None` vs `Partial`) **measures** the delta.
- **R8 not enabled / misconfigured.** Shipping **debuggable/unshrunk** release builds (R8 off) leaves the APK **large** and **unoptimized**; misconfigured **keep rules** either break the app (over-shrinking reflection-used code) or defeat shrinking (keeping everything). Need **R8 on** for release with correct `-keep` rules.
- **Obfuscation vs debuggability (Ch 83).** R8 **obfuscates** (renames) — great for size/IP, but stack traces become unreadable unless you **retain + upload the mapping file** for **deobfuscation** (Crashlytics, Ch 83). Forgetting the mapping file makes crashes undebuggable.

- **Resource shrinking.** Beyond code, `shrinkResources` removes unused resources — more size savings, must be paired with R8 code shrinking.
- **Measuring the win.** Claims of "faster" must be **measured** — **Macrobenchmark** `StartupTimingMetric` / `FrameTimingMetric` with `CompilationMode.None` (no profile) vs `Partial` (with profile) quantifies the baseline-profile gain; APK analyzer quantifies R8 size savings; field vitals confirm (Ch 85).

The architecture: ship a **Baseline Profile** covering **launch + the critical journeys** (feed scroll, key screens) — generated with the **Baseline Profile Gradle plugin** (Macrobenchmark- based generator) and packaged so ART **AOT-compiles** that code **at install** (fast first launch, Ch 88; smooth first scroll, Ch 90); enable **R8** for release (**shrink + optimize + obfuscate**, smaller/faster), with correct `-keep` **rules** and `shrinkResources`, and **retain + upload the mapping file** for crash **deobfuscation** (Ch 83); and **verify** everything with **Macrobenchmark** (`None` vs `Partial`) and **field** metrics (Ch 85). This chapter is about **fast-from-first-launch + smaller + verified**.

Three practical use cases

1. **Baseline Profile for StreamCart launch + feed (Ch 88/90).** Generate a profile covering app launch and the home-feed scroll (Ch 71); Macrobenchmark shows cold start **~25% faster** (`None` vs `Partial`) and first-scroll jank cut — verified before shipping.
2. **R8 shrink + resource shrink.** Enable R8 + `shrinkResources` for release; APK drops significantly (dead code/resources removed), with `-keep` rules for reflection-used (Gson/Hilt/serialization) classes; mapping file uploaded for Crashlytics (Ch 83).
3. **Verify the profile applied.** Confirm via `ProfileInstaller` + Macrobenchmark that `CompilationMode.Partial` reflects the shipped profile — not silently missing.

Why this is architectural

Baseline Profiles + R8 compose cold start (Ch 88), jank (Ch 90), Macrobenchmark measurement (Ch 85), crash deobfuscation (Ch 83), and release (Ch 41). The decisions — profile the right journeys, enable R8 with correct keep rules, retain the mapping file, verify with Macrobenchmark — determine whether the app is fast from first launch and small, without breaking crash diagnosability.

92.2 Requirements Engineering

Functional requirements

ID	Requirement
F1	Ship a Baseline Profile covering launch + critical journeys (feed scroll, key screens).
F2	Generate it with the Baseline Profile Gradle plugin (Macrobenchmark generator).
F3	Enable R8 for release: shrink + optimize + obfuscate; `shrinkResources`.
F4	Correct `-keep` rules (reflection/serialization/DI) — don't over- or under-shrink.
F5	Retain + upload the mapping file for crash deobfuscation (Ch 83).
F6	Verify with Macrobenchmark (`None` vs `Partial`) and field metrics (Ch 85).

Non-functional requirements

Attribute	Policy
First-launch speed	AOT startup path; fast from install (Ch 88).
First-scroll smoothness	Scroll path in profile (Ch 90).
Size	R8 + resource shrinking; smaller APK.
Debuggability	Mapping file retained/uploaded (Ch 83).
Verified	Macrobenchmark `None` vs `Partial` (Ch 85).

Scope

- **In scope:** AOT vs JIT vs interpreted, Baseline Profiles (generation via Gradle plugin/ Macrobenchmark, packaging, `ProfileInstaller`, coverage = launch + journeys), effect on cold start (Ch 88) + first-scroll jank (Ch 90), R8 (shrink/optimize/obfuscate), `-keep` rules, `shrinkResources`, mapping file + deobfuscation (Ch 83), Cloud/Startup profiles context, verification (Macrobenchmark `None` / `Partial`, Ch 85), APK size analysis.
- **Out of scope:** cold-start optimization detail (Ch 88), jank detail (Ch 90), Macrobenchmark setup detail (Ch 85), crash reporting (Ch 83), release mechanics (Ch 41).

Assumptions & constraints

- AGP with the **Baseline Profile Gradle plugin** + **Macrobenchmark** module (Ch 85); **R8** enabled for release (`minifyEnabled true`, `shrinkResources true`); `ProfileInstaller` packaged. Profile covers **launch + key journeys**; mapping file **retained + uploaded** (Ch 83). Verified with `CompilationMode.None` vs `Partial`. `minSdk 24` (ART; Baseline Profiles supported broadly via `ProfileInstaller`).

92.3 System Design Fundamentals

- **Interpreted → JIT → AOT (why the first run is slow).** ART runs DEX bytecode **interpreted** at first; it **JIT-compiles** frequently-run ("hot") methods to native over time (and writes a **profile** of what was hot, used to AOT-compile later in the background). So **fresh installs** are slow until JIT/profile-guided compilation warms up. **AOT** (ahead-of-time) compilation **before** running avoids this. A **Baseline Profile** is a way to get **AOT compilation of the code you choose, at install time** — skipping the slow warm-up for the paths that matter.
- **Baseline Profiles — what they are.** A **Baseline Profile** is a list of **classes/methods** (in a human-readable rules format, compiled to a binary) that you **ship with the app**; at **install** (and app update), ART **AOT-compiles** exactly those → the **startup + critical-path code is native from the first launch**. Typical wins: **~20–30% faster cold start** (Ch 88) and **smoother first scroll** (Ch 90). It's additive to ART's own profile-guided compilation (which still kicks in over time for the rest).
- **Generating a Baseline Profile.** Use the **Baseline Profile Gradle plugin** with a **Macrobenchmark**-based **generator** that **exercises the critical journeys** (launch, scroll the feed, open key screens) on a device/emulator; it **records** the methods/classes those journeys touch into the profile. **Coverage matters:** the generator must drive the **real most-used journeys** — a profile covering only launch under-delivers on scroll (Ch 90). Include **launch + the top user journeys**.
- **Packaging & applying (`ProfileInstaller`).** The profile is **packaged** into the app (AAB/APK) and **applied** at install (Play) or via `ProfileInstaller` (bundled library) on first run for sideloads/older setups. **Verify** it's applied — a profile that isn't packaged/ applied gives **no** benefit. **Cloud Profiles** / **Startup Profiles** are related mechanisms (Play-aggregated field profiles; profiles that also affect DEX layout) — the Baseline Profile plugin is the standard app-shipped approach.
- **Effect on cold start (Ch 88) & jank (Ch 90).** Baseline Profiles directly improve the two most visible perf dimensions: **cold start** (AOT startup path — faster first launch, Ch 88) and **first-scroll jank** (AOT Compose/scroll path — smoother first scroll, Ch 90). These chapters **reference this one** for the AOT mechanism.
- **R8 — shrink, optimize, obfuscate. R8** (the default release compiler/shrinker) does: **shrinking** (remove unused classes/methods/fields — **tree-shaking** from entry points), **optimization** (inlining, dead-code elimination, class merging), and **obfuscation** (rename to short names — smaller + harder to reverse-engineer). Result: **smaller DEX/APK**, **faster to load/verify**, **less to compile/init**. Enable with `minifyEnabled true` for release; pair with `shrinkResources true` to drop unused resources.
- **`-keep` rules (the correctness knob).** R8 shrinks from **reachable entry points**; code reached only via **reflection/serialization/JNI** (Gson/Moshi models, Hilt/Dagger, Compose stability sometimes, `Parcelable`, enums used by name) may be **wrongly removed/renamed** → runtime crash. `-keep` **rules** (ProGuard/R8 rules) tell R8 to **keep** those. **Too few** → crashes; **too many** (`-keep class ** { *; }`) → defeats shrinking. Use **library-provided consumer rules** + **minimal targeted keeps**; test the **release** build.
- **Obfuscation vs crash debuggability (Ch 83).** Obfuscation renames everything, so **stack traces are gibberish** unless you **retain the** `mapping.txt` and **upload it** (to Crashlytics/ Play) for **deobfuscation/symbolication** (Ch 83). **Always keep and upload the mapping file** per release — otherwise production crashes are undebuggable.
- **Verify with Macrobenchmark (Ch 85).** Prove the wins: **Macrobenchmark** with `CompilationMode.None` (no AOT — simulates JIT-cold) vs `CompilationMode.Partial` (with baseline profile) **quantifies** the baseline-profile gain on **startup** (`StartupTimingMetric`, Ch 88) and **frames** (`FrameTimingMetric`, Ch 90).

> **APK Analyzer** quantifies R8 size savings. **Field vitals** (Ch 85) confirm in production. **Measure — don't assume.**

Core insight: **make the app fast from the very first launch and smaller by shipping a Baseline Profile and enabling R8: a Baseline Profile lists the startup + critical-journey classes/methods so ART AOT-compiles them at install time instead of leaving them interpreted/JIT-cold, which is why fresh installs are otherwise slow — generate it with the Baseline Profile Gradle plugin driving a Macrobenchmark that exercises the real most-used journeys (launch, feed scroll, key screens — coverage matters, a launch-only profile under-delivers on scroll), package it via ProfileInstaller, and it directly improves cold start (~20–30%, Ch 88) and first-scroll jank (Ch 90); R8 separately shrinks (tree-shakes dead code from entry points), optimizes (inlining/ dead-code elimination), and obfuscates (renames) the release build for a smaller, faster-loading APK, enabled with minifyEnabled + shrinkResources and correct -keep rules for reflection/ serialization/DI code (too few keeps crashes at runtime, too many defeats shrinking) — but obfuscation makes crash stack traces unreadable, so you must retain and upload the mapping.txt for deobfuscation (Ch 83); and you verify both wins with Macrobenchmark (CompilationMode.None vs Partial for the profile's startup/frame delta) and the APK Analyzer and field vitals (Ch 85), because these are the highest-impact, lowest-effort perf wins only when actually applied and measured.**

92.4 High-Level Architecture


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Components & ownership

- **R8** shrinks/optimizes/obfuscates release → smaller/faster + `mapping.txt` (upload for deobfuscation, Ch 83).
- **Baseline Profile** (generated via Gradle plugin + Macrobenchmark) → **AOT** startup + journeys at install → **fast first launch** (Ch 88) + **smooth first scroll** (Ch 90).
- **Verify** with Macrobenchmark (`None` vs `Partial`) + APK Analyzer + field (Ch 85).

Direction: code → R8 (smaller, +mapping) + Baseline Profile (AOT via ProfileInstaller) → fast-from-first-launch + small → verified.

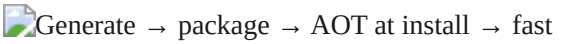
92.5 Detailed Component Design


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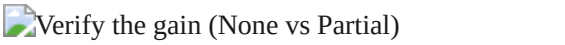
Component	Responsibility
`BaselineProfile`	startup + journey rules; generated (Macrobenchmark), applied (ProfileInstaller)
`R8Config`	`minifyEnabled` / `shrinkResources` , keep rules, mapping file (Ch 83)
`Verification`	Macrobenchmark `None` vs `Partial` (Ch 88/90), APK size, field (Ch 85)

92.6 End-to-End Data Flow

Generate → package → AOT at install → fast



Verify the gain (None vs Partial)



92.7 Complete Kotlin / Gradle Implementation

Package/module: `:baselineprofile`, app `build.gradle`.

92.7.1 Baseline Profile generator (Macrobenchmark, Ch 85)

```
@RunWith(AndroidJUnit4::class)
class BaselineProfileGenerator {
    @get:Rule val rule = BaselineProfileRule()
    @Test fun generate() = rule.collect(packageName = "com.streamcart") {
        pressHome(); startActivityAndWait()                      // LAUNCH path (Ch 88)
        // exercise the REAL critical journeys so the profile covers them (coverage matters):
        device.findObject(By.res("feed")).fling(Direction.DOWN)   // FEED SCROLL path (Ch 90)
        device.findObject(By.res("product_0")).click(); device.pressBack()   // key screen
    }
    // Output: baseline-prof.txt (classes/methods) -> packaged; ART AOT-compiles them at
install.
}
```

92.7.2 R8 + resource shrinking + mapping (app build.gradle)

```
android {
    buildTypes {
        release {
            isMinifyEnabled = true            // R8: shrink + optimize + obfuscate
            isShrinkResources = true          // drop unused resources (pair with R8)
            proguardFiles(
                getDefaultProguardFile("proguard-android-optimize.txt"),
                "proguard-rules.pro",          // our -keep rules
            )
            // mapping.txt is produced automatically -> upload for deobfuscation (Ch 83)
        }
    }
}
// Baseline Profile Gradle plugin wires the generated profile into the release build.
```

92.7.3 Keep rules (proguard-rules.pro) — correctness

```
# Keep reflection/serialization-used models (Gson/Moshi) - else R8 removes/renames -> runtime
crash
-keep class com.streamcart.model.** { *; }
-keepclassmembers class com.streamcart.model.** { <fields>; }
# Hilt/Dagger, Parcelable, enums-by-name usually covered by library consumer rules; add targeted
keeps only as needed.
# AVOID: -keep class ** { *; }  // keeps everything -> defeats shrinking
# Keep source file + line numbers for readable (deobfuscated) stack traces (Ch 83):
-keepattributes SourceFile,LineNumberTable
-renamesourcefileattribute SourceFile
```

92.7.4 Verify with Macrobenchmark (None vs Partial, Ch 85/88/90)

```
@RunWith(AndroidJUnit4::class)
class ProfileVerificationBenchmark {
    @get:Rule val rule = MacrobenchmarkRule()
    private fun startup(mode: CompilationMode) = rule.measureRepeated(
        packageName = "com.streamcart", metrics = listOf(StartupTimingMetric()),  // Ch 88
        compilationMode = mode, startupMode = StartupMode.COLD, iterations = 10,
    ) { pressHome(); startActivityAndWait() }
    @Test fun noProfile() = startup(CompilationMode.None())        // JIT-cold baseline
    @Test fun withProfile() = startup(CompilationMode.Partial())   // with baseline profile ->
faster
    // Delta (None - Partial) = the baseline-profile win. Also FrameTimingMetric for scroll (Ch
90).
}
```

92.7.5 Upload mapping for deobfuscation (Ch 83)

```
// Crashlytics Gradle plugin uploads mapping.txt automatically on release build;
// or upload to Play Console. Without it, obfuscated crash stack traces are unreadable (Ch 83).
// firebaseCrashlytics { mappingFileUploadEnabled = true }   // per release
```

92.7.6 Module structure

```
:baselineprofile  BaselineProfileGenerator (Macrobenchmark, Ch 85) -> baseline-prof.txt
:app  R8 (minify + shrinkResources) + keep rules + baseline profile plugin + mapping upload (Ch
83)
:benchmark  ProfileVerificationBenchmark (None vs Partial, Ch 88/90)
# Fast first launch -> Ch 88; smooth first scroll -> Ch 90; verify -> Macrobenchmark (Ch 85).
```

92.8 Concurrency and Race Conditions

> Baseline Profiles & R8 are **build/compilation-time** concerns, so classic runtime races don't apply. The analogous pitfalls are **correctness/coverage** issues:

Scenario	Risk	Handling
Profile covers only launch	under-delivers on scroll	exercise real journeys (Ch 90)
Profile not applied	no benefit	ProfileInstaller + verify (None vs Partial)
Over-shrinking (too few keeps)	runtime crash	keep reflection/serialization/DI classes
Under-shrinking (keep all)	no size win	minimal targeted keeps
Mapping file lost	undebuggable crashes	retain + upload mapping.txt (Ch 83)
Release untested (R8 on)	ships broken	test the release (minified) build
Assuming the win	unverified	Macrobenchmark None vs Partial (Ch 85)

- **Coverage is the baseline-profile analog of correctness:** a profile only helps the journeys it **covers**, so the generator must drive the **real** critical paths (launch **and** scroll **and** key screens). Verify with Macrobenchmark that `Partial` beats `None` for those journeys.
- **Keep rules are the R8 correctness knob:** R8's static reachability can't see **reflection/ serialization**, so missing keeps cause **release-only** crashes (invisible in debug). Always **test the minified release build**, and lean on **library consumer rules** plus **minimal** targeted keeps.
- **The mapping file is non-negotiable for debuggability (Ch 83):** obfuscation + no mapping = undebuggable production crashes — retain and upload it **every release**.

92.9 Failure Scenarios and Recovery

Failure	Detection	Response	Outcome
Slow first launch	Macrobenchmark None vs Partial (Ch 88)	ship Baseline Profile	fast first launch
First-scroll jank	FrameTimingMetric (Ch 90)	include scroll path in profile	smooth
Profile not applied	Partial == None	fix packaging/ProfileInstaller + verify	applied
Release crash (R8)	crash on reflection class	add targeted -keep rule	stable
Unreadable crashes	obfuscated stack trace (Ch 83)	retain + upload mapping.txt	deobfuscated
Large APK	APK Analyzer	R8 + shrinkResources + minimal keeps	smaller
Unverified win	no numbers	Macrobenchmark + APK Analyzer (Ch 85)	measured


92.9 Failure Scenarios and Recovery

> **Ship the profile, enable R8, keep the mapping file — and verify.** Baseline Profiles + R8 are the highest-impact, lowest-effort perf wins, but only when the profile **covers the real journeys** and is **actually applied** (verify None vs Partial, Ch 85), R8 has **correct keep rules** (test the release build), and the **mapping file is uploaded** so crashes stay debuggable (Ch 83). Measure the gain — don't assume it.

92.10 Scaling the Implementation

- **Level 1.** Enable R8 (`minifyEnabled`) + `shrinkResources` for release; upload the mapping file (Ch 83); add library consumer keep rules.
- **Level 2.** Add a Baseline Profile (Gradle plugin) covering **launch**; verify cold start with Macrobenchmark `None` vs `Partial` (Ch 88).
- **Level 3 — production.** Expand the profile to **key journeys** (feed scroll, Ch 90); CI verification of the profile delta; APK-size tracking; field vitals confirmation (Ch 85).
- **Level 4 — large.** Per-flow profiles for all major journeys; automated profile regeneration in CI; size budgets; R8 optimization passes tuned; dynamic feature modules.
- **Level 5 — enterprise.** Profile + size governance, Cloud/Startup Profiles, per-module keep- rule hygiene, automated startup/size regression gates (Ch 41), cross-release AOT/size dashboards (Ch 85).

Avoid overengineering: R8 + `shrinkResources` + a Baseline Profile covering launch + the top journey, mapping file uploaded, verified with Macrobenchmark, captures most of the win. Don't ship without a profile; don't keep everything; don't forget the mapping file; don't claim wins you didn't measure.

92.11 Alternative Approaches and Trade-offs

Dimension	Option A	Option B	When
First-run speed	Baseline Profile (AOT)	JIT-only	Baseline Profile (fast first launch)
Profile coverage	Launch + journeys	launch only	Journeys (smooth scroll too, Ch 90)
Release build	R8 on (shrink/optimize)	R8 off	R8 on (smaller/faster)
Keep rules	Minimal targeted	keep-all / none	Minimal targeted (shrink + correct)
Resources	`shrinkResources`	keep all	Shrink (size)
Debuggability	Upload mapping.txt (Ch 83)	no mapping	Upload (readable crashes)
Proof	Macrobenchmark (Ch 85)	assume	Measure

Baseline Profile vs JIT-only: without a profile, startup + scroll run **interpreted/JIT** on first launches — slow/janky until warm. A **Baseline Profile AOT-compiles** those paths at install → **fast/smooth from the first launch** (~20–30% startup, Ch 88/90). It's additive to ART's own profiling. **Always ship one** covering launch + top journeys.

R8 on vs off: shipping **unshrunk** release builds wastes **size** (dead code/resources) and **load/compile time**. **R8** tree-shakes, optimizes, and obfuscates → **smaller, faster**. The cost is **keep-rule** maintenance and **mapping-file** management — both well-trodden. **Always enable R8** for release with correct keeps + mapping upload.

Minimal targeted keeps vs keep-all: `-keep class ** { *; }` is "safe" but **defeats shrinking** (no size win). **Too few** keeps crashes on reflection/serialization. **Minimal, targeted** keeps (library consumer rules + specific models) **shrink effectively while staying correct** — verified by testing the **release** build. Prefer minimal targeted keeps.

92.12 Security & Privacy Considerations

- **Obfuscation raises the reverse-engineering bar (not a security guarantee):** R8 renaming makes the app **harder to reverse-engineer** (IP/tamper resistance), but it is **not encryption** — don't rely on obfuscation to hide **secrets** (keys still shouldn't be in the APK, Ch 24).
- **Mapping file is sensitive (Ch 83):** `mapping.txt` **de-obfuscates** your app — treat it as **internal**, upload it to your crash tooling securely (Ch 83), don't ship/leak it publicly.
- **Baseline profiles contain method/class names, not data:** compilation hints — not secrets or PII (Ch 28).
- **Shrinking can remove debug/log code:** R8 can strip debug logging in release (good for leak- prevention), but ensure **security-relevant** code isn't stripped by a wrong keep (Ch 24).
- **Resource shrinking + size** reduces attack surface marginally (less code) — not a primary security control.
- Build-time concern: these are **build** features; the security notes are **mapping-file confidentiality** (Ch 83) and **not trusting obfuscation as secret-hiding** (Ch 24).

92.13 Performance and Resource Optimization

- **Baseline Profiles = first-launch + first-scroll speed (Ch 88/90)** — AOT the startup + top journeys; the single biggest low-effort startup/jank win.
- **R8 = size + load/compile speed** — shrink + optimize + obfuscate; `shrinkResources` for assets.
- **Coverage drives the profile win** — profile the **real** journeys, not just launch (Ch 90).
- **Minimal keeps maximize shrinking** — keep only reflection/serialization/DI code.
- **Verify, quantify** — Macrobenchmark `None` vs `Partial` (startup/frames), APK Analyzer (size), field vitals (Ch 85).
- **Profiling:** cold start `None` vs `Partial` (Ch 88), frame timing `None` vs `Partial` (Ch 90), APK/DEX size before/after R8, resource count, field startup/jank (Ch 85).

92.14 Testing Strategy

The profile delta and release correctness are the things to test; size is measured.

```
class ProfileVerificationBenchmark {   // quantify the baseline-profile win (Ch 85/88/90)
    @get:Rule val rule = MacrobenchmarkRule()
    @Test fun startup_none() = measure(CompilationMode.None())        // JIT-cold baseline
    @Test fun startup_partial() = measure(CompilationMode.Partial())  // with profile -> assert
faster
    private fun measure(mode: CompilationMode) = rule.measureRepeated(
        packageName = "com.streamcart", metrics = listOf(StartupTimingMetric()),
        compilationMode = mode, startupMode = StartupMode.COLD, iterations = 10) { pressHome();
startActivityAndWait() }
}

// Release correctness (R8 keep rules):
//  - Run instrumented/smoke tests on the MINIFIED release build - catch over-shrunk reflection
classes.
//  - Verify Gson/Moshi serialization, Hilt graph, Parcelable, enums-by-name still work post-R8.
//  - Confirm mapping.txt produced + uploaded (Ch 83); deobfuscate a sample stack trace.
// Size: APK Analyzer before/after R8 + shrinkResources.
```

What to test: Macrobenchmark `None` vs `Partial` startup (Ch 88) + frames (Ch 90) — profile applied + faster; **release (minified) build** smoke/integration tests — keep rules correct (no reflection crash); mapping file produced/uploaded + sample deobfuscation (Ch 83); APK size delta (APK Analyzer). Field confirmation via vitals (Ch 85).

92.15 Production Observability

- **Metrics: cold start** p50/p95 (field vitals + Macrobenchmark `None` vs `Partial`, Ch 88/ 85), **first-scroll jank** (`FrameTimingMetric` `None` vs `Partial`, Ch 90), **APK/DEX size** per release, **crash deobfuscation** success (mapping uploaded?, Ch 83), profile-applied rate (ProfileInstaller).
- **Logs:** build-time R8/profile reports; ProfileInstaller status — no PII (Ch 28).
- **Debugging:** no startup win → profile applied? (`None` vs `Partial` equal → packaging issue); release crash → missing `-keep` (test minified build); unreadable crashes → mapping not uploaded (Ch 83); large APK → R8/shrinkResources off or keep-all; scroll still janky → profile missing the scroll journey (Ch 90).
- **Alerting:** cold-start/jank regression release-over-release (profile broke?, Ch 88/90/41); APK size regression; deobfuscation failures (missing mapping, Ch 83).
- **Dashboards:** cold start + first-scroll (profile effect) by release; APK size trend; profile- applied rate; crash deobfuscation health.

92.16 System Design Interview Preparation

> ≥25 questions.

Fundamentals

1. *Interpreted/JIT/AOT?* Interpret first → JIT hot methods → AOT ahead-of-time.
2. *Why first launch slow?* Code is interpreted/JIT-cold until warm.
3. *Baseline Profile?* Ship class/method list → ART AOT-compiles at install.
4. *Typical win?* ~20–30% faster cold start (Ch 88) + smoother first scroll (Ch 90).
5. *R8?* Shrink + optimize + obfuscate the release build.

Baseline Profiles 6. *Generate how?* Baseline Profile Gradle plugin + Macrobenchmark generator. 7. *Coverage?* Launch + real critical journeys (not launch-only). 8. *Apply how?* Packaged + ProfileInstaller (install time). 9. *Effect?* Faster cold start (Ch 88), smoother first scroll (Ch 90). 10. *Cloud/Startup Profiles?* Play-aggregated / DEX-layout variants.

R8 11. *Shrinking?* Tree-shake dead code from entry points. 12. *Optimization?* Inlining, dead-code elimination, class merging. 13. *Obfuscation?* Rename to short names — smaller + harder to RE. 14. *Enable?* `minifyEnabled true` (release). 15. *shrinkResources?* Remove unused resources.

Keep rules 16. *Why needed?* R8 can't see reflection/serialization/JNI. 17. *Too few?* Runtime crash (removed/renamed class). 18. *Too many?* Defeats shrinking. 19. *Best practice?* Library consumer rules + minimal targeted keeps. 20. *Test?* On the minified release build.

Obfuscation/debuggability 21. *Crash stack traces?* Obfuscated gibberish without mapping. 22. *Fix?* Retain + upload mapping.txt (Ch 83). 23. *Mapping sensitive?* Yes — internal only.

Verify 24. *Measure profile win?* Macrobenchmark `None` vs `Partial` (Ch 85). 25. *Measure size?* APK Analyzer. 26. *Confirm in field?* Vitals cold start/jank (Ch 85).

Design-change challenges

1. **Make first launch faster.** Baseline Profile covering launch; verify None vs Partial (Ch 88).
2. **Smooth first scroll.** Include the scroll journey in the profile (Ch 90).
3. **Shrink the APK.** R8 + shrinkResources + minimal keeps.
4. **Keep crashes debuggable.** Retain + upload mapping.txt (Ch 83).
5. **Prove the win.** Macrobenchmark None vs Partial + APK Analyzer (Ch 85).

Mock walkthrough (abridged)

Clarify (first-launch speed? size? journeys?) → AOT vs JIT (why first run slow) → Baseline Profile via Gradle plugin + Macrobenchmark generator covering launch + journeys (Ch 88/90) → package (ProfileInstaller) → R8 (minify + shrinkResources) + correct keep rules → retain + upload mapping.txt (Ch 83) → verify Macrobenchmark `None` vs `Partial` + APK Analyzer + field vitals (Ch 85) → observability (cold start/size/deobfuscation by release).

92.17 Practical Exercises

1. **Beginner.** Enable R8 + `shrinkResources` for release, add library keep rules, and confirm the mapping file is uploaded for Crashlytics (Ch 83). *Eval:* smaller APK; readable crashes.

2. **Intermediate.** Add a Baseline Profile (Gradle plugin) covering app launch and verify the cold-start delta with Macrobenchmark `None` vs `Partial` (Ch 88). *Eval:* measurable startup win.
3. **Advanced.** Expand the profile to cover feed scroll (Ch 90) and verify the frame delta (`FrameTimingMetric` `None` vs `Partial`); tighten keep rules to maximize shrinking. *Eval:* smoother first scroll; smaller APK; no release crash.
4. **Production debugging.** Diagnose "no startup win" (profile not applied — fix packaging) and a release-only reflection crash (add targeted `-keep`). *Eval:* profile applies; crash fixed.
5. **System design challenge.** Design CI gates for cold-start (Ch 88), frame timing (Ch 90), and APK size, plus mapping-file upload verification (Ch 83). *Eval:* regressions gated; crashes debuggable.

(Solutions withheld unless requested.)

92.18 Chapter Summary & Reference Sheet

Key decisions: make the app **fast from the very first launch** and **smaller** by shipping a **Baseline Profile** and enabling **R8**: a **Baseline Profile** lists the **startup + critical- journey classes/methods** so **ART AOT-compiles them at install** instead of leaving them **interpreted/JIT-cold** (the reason fresh installs are otherwise slow) — **generate** it with the **Baseline Profile Gradle plugin** driving a **Macrobenchmark** that exercises the **real most-used journeys** (launch, feed scroll, key screens — **coverage matters**, a launch-only profile under- delivers on scroll), **package** it via **ProfileInstaller**, and it directly improves **cold start** (~**20–30%**, Ch 88) and **first-scroll jank** (Ch 90); **R8** separately **shrinks** (tree-shakes dead code), **optimizes** (inlining/dead-code elimination), and **obfuscates** (renames) the release build for a **smaller, faster-loading APK**, enabled with ** `minifyEnabled`

- `shrinkResources` ** and **correct** `-keep` **rules** for reflection/serialization/DI code (**too few keeps crashes** at runtime, **too many defeats shrinking**) — but obfuscation makes crash stack traces **unreadable**, so you must **retain and upload the** `mapping.txt` for **deobfuscation** (Ch 83); and you **verify** both wins with **Macrobenchmark** (`CompilationMode.None` **vs** `Partial` for the profile's startup/frame delta), the **APK Analyzer** (size), and **field vitals** (Ch 85) — because these are the **highest-impact, lowest- effort** perf wins **only when actually applied and measured**.

Concepts: interpreted/JIT/AOT, Baseline Profiles (generation via Gradle plugin + Macrobenchmark, coverage, ProfileInstaller), effect on cold start (Ch 88) + first-scroll jank (Ch 90), R8 (shrink/optimize/obfuscate), `-keep` rules, `shrinkResources`, `mapping.txt` + deobfuscation (Ch 83), verification (`None` vs `Partial`, APK Analyzer, Ch 85).

Common mistakes: no Baseline Profile (JIT-cold first launch); profile covering only launch (no scroll win, Ch 90); profile not applied (unverified); R8 off (large/slow); too few keeps (release crash); keep-all (no shrink); forgetting the mapping file (undebuggable crashes, Ch 83); assuming the win without Macrobenchmark (Ch 85); not testing the minified release build.

Checklist: ☐ Baseline Profile covering launch + top journeys (feed scroll, Ch 90) ☐ generated via Gradle plugin + Macrobenchmark; applied via ProfileInstaller ☐ verify `None` vs `Partial` startup (Ch 88) + frames (Ch 90) ☐ R8 on (`minifyEnabled`) + `shrinkResources` ☐ minimal targeted `-keep` rules (reflection/serialization/DI) ☐ test the minified release build ☐ retain + upload `mapping.txt` for deobfuscation (Ch 83) ☐ APK Analyzer size delta ☐ field vitals confirm (Ch 85) ☐ CI gates for startup/frames/size (Ch 41).

Key APIs/Gradle: `BaselineProfileRule` (generator) + Baseline Profile Gradle plugin, `ProfileInstaller`, `CompilationMode.None`/`Partial` + `StartupTimingMetric` (Ch 88)/ `FrameTimingMetric` (Ch 90) via `MacrobenchmarkRule` (Ch 85), `isMinifyEnabled`/`isShrinkResources` (R8), `proguard-rules.pro` (`-keep`, `-keepclassmembers`, `-keepattributes SourceFile,LineNumberTable`), `mapping.txt` upload (Crashlytics/Play, Ch 83), APK Analyzer, field vitals (Ch 85).

References:

- Baseline Profiles: https://developer.android.com/topic/performance/baselineprofiles/overview
- Create a Baseline Profile: https://developer.android.com/topic/performance/baselineprofiles/create-baselineprofile
- R8 / shrink, obfuscate, optimize: https://developer.android.com/build/shrink-code
- Deobfuscate crash stack traces: https://developer.android.com/build/shrink-code#decode-stack-trace
- (composes) Ch 88 Cold Start, Ch 90 Jank, Ch 85 Perf/Macrobenchmark, Ch 83 Crash/Deobfuscation, Ch 45 Startup, Ch 41 Release

Chapter review (Phase 4)

- **Completeness:** all 18 sections; architecture/class/sequence/None-vs-Partial/failure diagrams; generator + R8 config + keep rules + verification + mapping-upload code; concurrency (build-time → coverage/keeps/mapping pitfalls); failure (ship+R8+mapping+verify principle); 5 levels; profile-vs-JIT, R8-on-vs-off, keeps trade-offs; security (mapping confidential, obfuscation ≠ secret-hiding); perf (the whole chapter); None-vs-Partial + release-correctness + size tests; observability (startup/size/deobfuscation by release); 26 Q&A + 5 challenges. ✅
- **Expand later:** cold start (Ch 88), jank (Ch 90), Macrobenchmark (Ch 85), crash/deobfuscation (Ch 83).
- **Interview coverage:** exceeds ≥25 bar.